HYBRID PARTICIPATION INNOVATION LAB · ARTICLE 1

Transforming Risk Perception and Fostering Cooperatively the Agri-PhotoVoltaics Technology With Farmers in Five Countries

Antonucci, Volterrani, Serra, Cornaro, Petitta & Bovesecchi (2025) · AgriVoltaics Conference Proceedings 3 · REGACE project

Presented by Next Wave

Mahsa Shahi · Kuhyar Saeedi · Farida Safdari · Gelila Tsehaye Melesse · Md Nazmul Islam · Pranto Argha Sarker

THE PAPER AT A GLANCE

How do farmers see agrivoltaics, and can they help shape it?

Research question

How do farming communities perceive agrivoltaics, and which social, economic and practical factors shape acceptance?

Who and where

The sociological work package (WP6) of the EU project REGACE, with farmers in Italy, Austria, Germany, Israel and Greece.

Method

Mixed methods: 44 in-depth interviews, then participatory sessions combining Open Space, World Café and a fishbowl.

Core argument

A disruptive technology only works if end users take part in shaping it from the first stages, not just at the end.

THE TECHNOLOGY AND THE GAP

Solar panels and crops on the same land

What REGACE builds

Next-generation agrivoltaic greenhouses: photovoltaic panels combined with farming, plus CO2 enrichment for higher yields.

The promise: energy security and food security from the same land.

What research was missing

Most studies measure technical performance. Farmers' perceptions, concerns and needs were mostly ignored.

The authors call this a major barrier to adoption.

METHOD

Mixed methods, built for diversity

44
in-depth interviews

5
countries

8–15
people per session

30–50
minutes per interview

Who was chosen (purposive and snowball sampling)

Different farm types and sizes, different experience with technology, different regions, and different social, religious and ethnic backgrounds. Interviews in the farmers' own languages.

How they kept it rigorous

Triangulation: interviews cross-checked with sessions.
Member checking: early findings shown back to participants.

HOW A PARTICIPATORY SESSION WORKED

From open agenda to structured dialogue

1 · INFORM

Scientific input

Technology partners explain the REGACE greenhouse.

2 · OPEN

Open Space

Farmers propose their own topics; breakout groups form around them.

3 · DEEPEN

World Café

Rotating tables answer targeted questions; hosts keep the thread; concept maps record ideas.

4 · RETURN

Fishbowl

Results are given back and discussed in a participatory restitution.

Why combine them? Open Space brings out themes freely; World Café refines them. Sessions repeat across the year to catch seasonal risks.

FIRST YEAR ON THE GROUND

What the team did, 2023–2024

When	Activity	Aim	Result
May 2023	International sociological training centre	Prepare partners for fieldwork	9 trained partners
Jul–Dec 2023	44 in-depth interviews in 5 countries	Understand farmers' views on PV and agrivoltaics	Generally positive attitude; main risks and opportunities
Jan–May 2024	2 World Cafés (Germany, Italy)	Mixed participatory sessions	Detailed view of risks, opportunities and preconditions

The paper reports only this **first year**; the project continues with farmers in later phases.

FINDINGS

Farmers say yes, but with conditions

Positive, if adapted

Openness to agrivoltaics with little difference between regions.

Acceptance depends on fitting local crops, climate and infrastructure.

Barriers

High upfront cost and uncertain return.

Complex maintenance; need for training, technical help and insurance.

Grid connection, licensing and bureaucracy.

Opportunities

Full-service contracts: due diligence, training, maintenance, updates.

Links with irrigation and precision farming.

Local energy communities to share benefits.

⚠ **Without policy reform, adoption risks being limited to well-resourced farmers, widening inequality.**

OUR CRITICAL READING

Strong on listening, less clear on what changed

Strengths

Five countries and diverse farmers, not one "public".

Methods in sequence: each one feeds the next.

Results returned to participants (member checking, fishbowl).

Limits

Small sample, first year only; the authors call for large surveys.

Selection effect: farmers who join may already be positive.

Perception more than co-design: the paper doesn't yet show what changed in the greenhouse because of farmers.

WHAT WE TAKE FOR OUR PROJECT

Lessons for our AI energy assistant

Borrow the sequence

Inform, Open Space, World Café, fishbowl: a tested path we can adapt for tenants.

Support, not just tech

Farmers asked for training and full service. Tenants will need the same: help, repairs, a person to call.

Watch inequality

Only well-resourced farmers may benefit. In our case: only digitally skilled tenants could.

Show the trace

We will document what tenants change in the assistant, the step this paper still leaves open.

FOR DISCUSSION

Questions for the class

1. If farmers had said no, could the greenhouse design have changed, or only its communication?

2. Do scientific briefings at the start empower participants, or frame the debate before it begins?

3. Which farmers were missing, and how could small farms be reached?